

CS/EE 120B Custom Laboratory

Project Report

Flappy Bird

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Introduction:

Flappy bird is a classic arcade game in which you try to navigate a bird through a loop of obstacles, for as long as possible. The game is displayed on a screen, with a different screen displaying the score, and instructions for navigation. Buttons will be used for navigation, as well as gameplay.

Elements of the project that may be buggy is the display of which the game is held on. When the bird passes one of the pipes, the game freezes for a slight bit. When 2 or more pipes appear on screen, the game seems to run slower, this is probably because of the poor refresh rate of the ST7735R.

Elements of Complexity

ST7735R 128*160 Pixel LCD Screen is used to display the general gameplay.

16*2 LCD Screen is used to display the general instructions, and the score accumulated during gameplay, as well as the high score achieved across total playtime, assuming not reset.

Arduino Uno EEPROM is used to save a high score across different startup times.

However, I did not successfully implement any form of music using the passive buzzer.

User Guide

The system starts up on the menu screen, with a blue background displayed on the ST7735. The game title, and the relevant instructions are then displayed on the 16*2 LCD. By clicking the leftmost button, as described on the 16*2 LCD, you are taken to the game, where it starts immediately upon release of the leftmost button. When the game is over, either through hitting the floor, or an obstacle, you are taken to the game over screen, where it displays your score, the high score accumulated, and relevant instructions. Clicking the left button allows you to replay the game. Clicking the right button returns to the menu. By clicking the right button in the menu, you then view the high score accumulated. By clicking the right button again, you go back to the menu. By holding the left button down for 2 seconds, you can reset the high score.

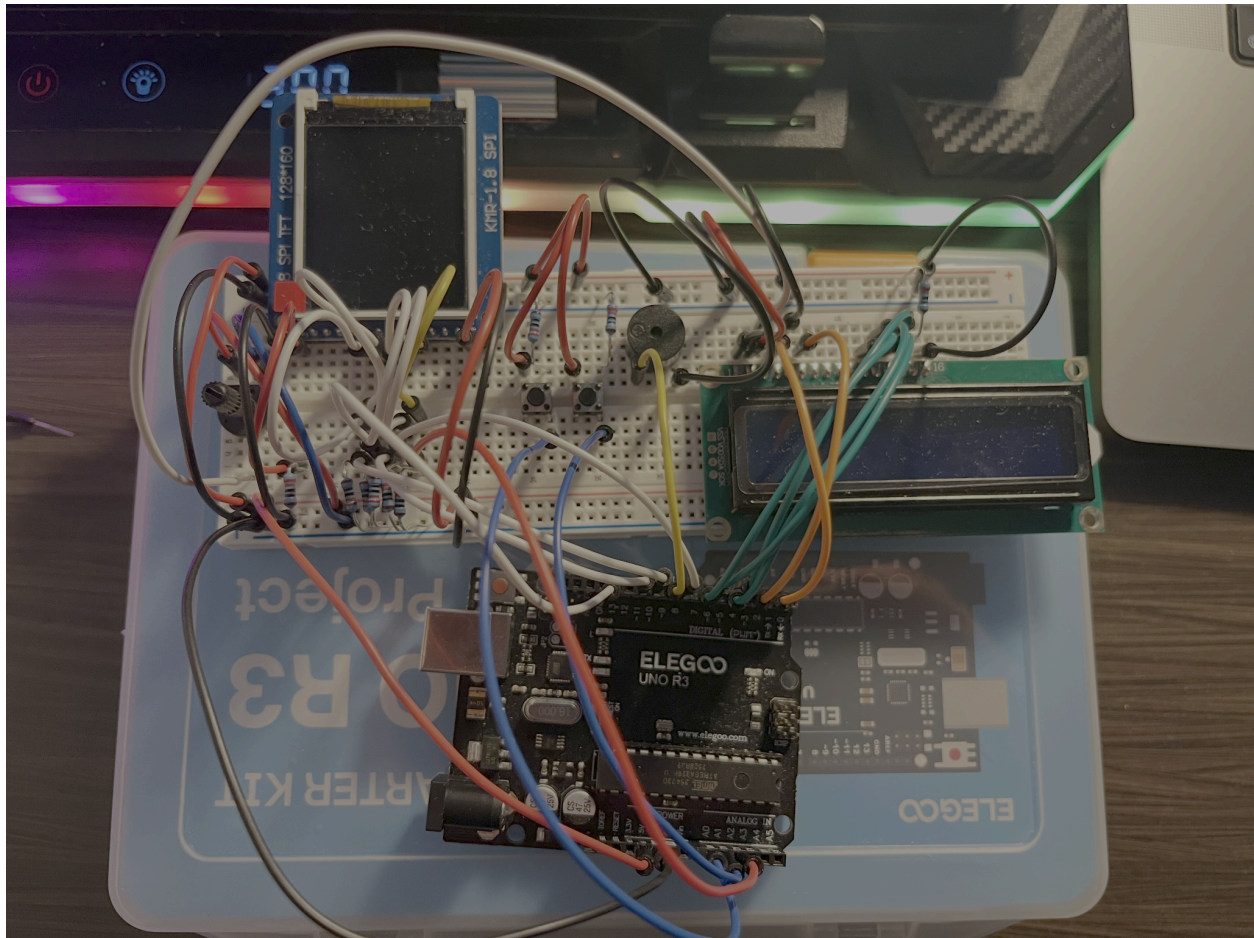
Hardware Components Used

- ST7735R 128*160 LCD Screen
- Arduino Uno
- 2x Buttons
- 1x Passive Buzzer
- 16*2 LCD Screen
- EEPROM

Software Libraries Used

No external software libraries were used in this program.

Wiring Diagram

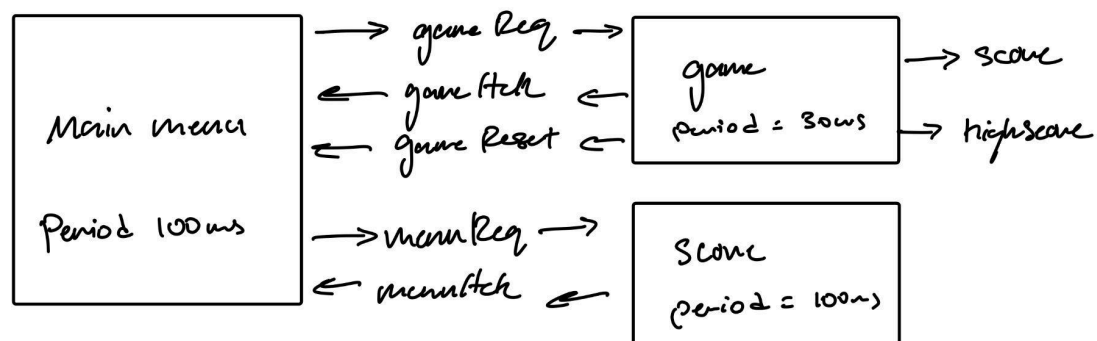


Task Diagram

unsigned char gameReq
unsigned char gameAck
unsigned char gameReset

unsigned char menuReq
unsigned char menuAck

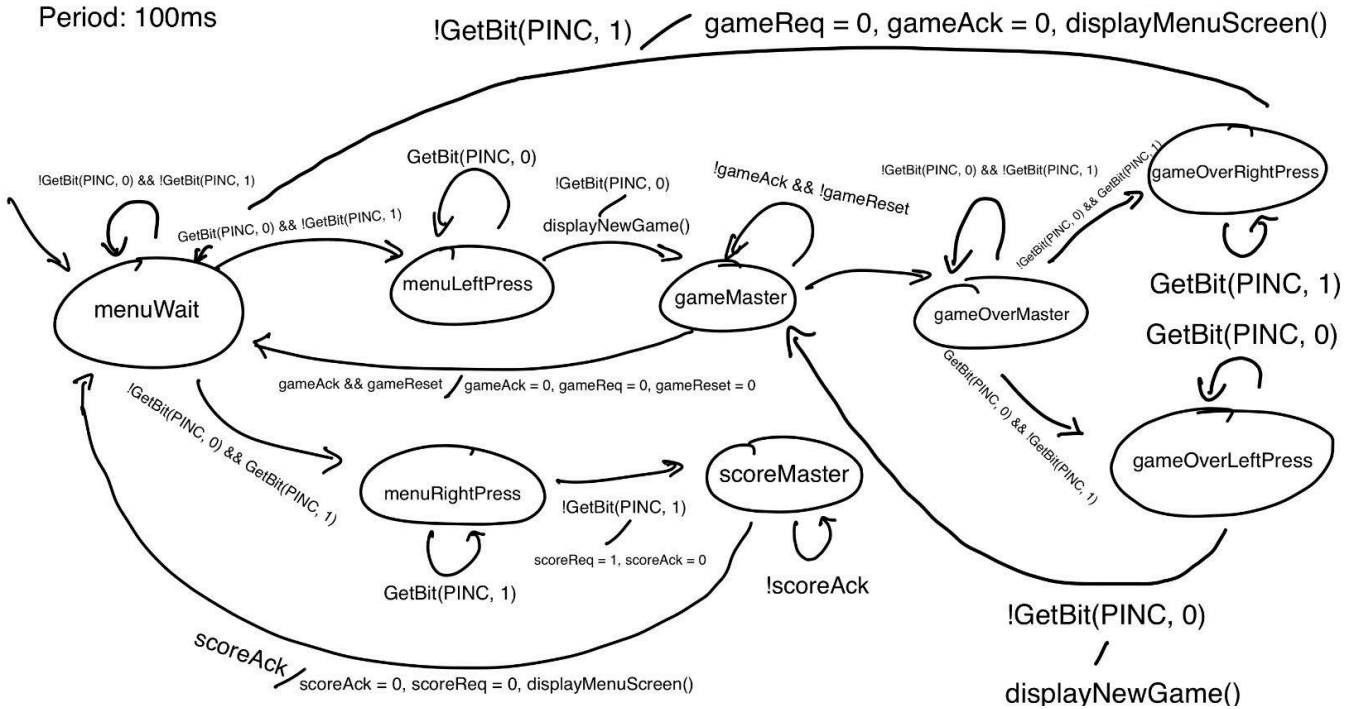
unsigned int score
unsigned int highScore



SynchSM Diagrams

Main Menu Tick:

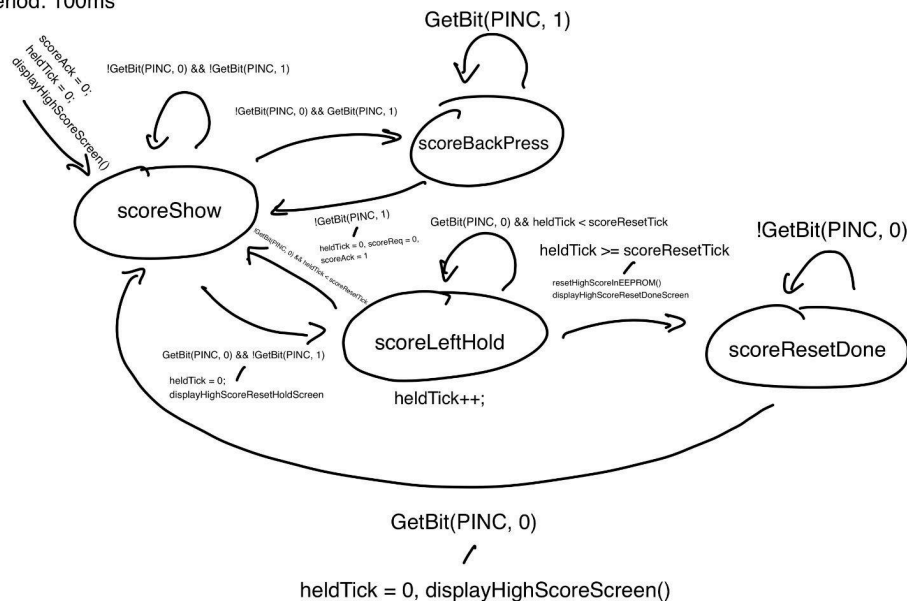
Period: 100ms



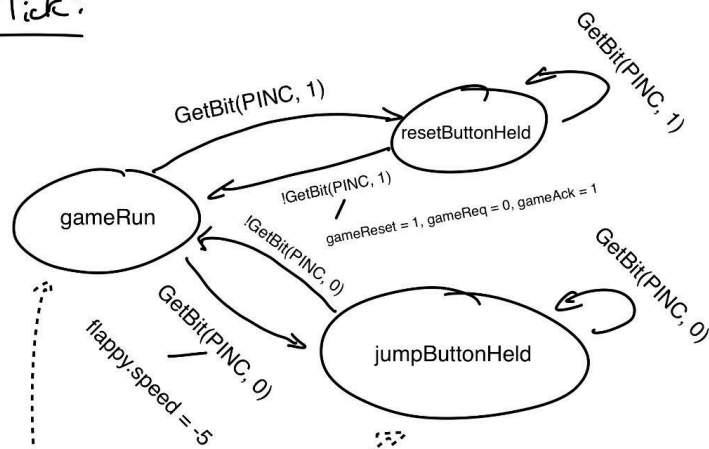
Score Tick:

static char heldTick = 0;

Period: 100ms



Game Tick:



- Draw over previous bird
- Update bird position
- Go through obstacle list
 - update position, based on score
 - Determine whether or not the bird has hit any of the pipes
 - check to see if you got a point by passing the pipe
 - check to see if pipe went offscreen, delete if so
- Add a new pipe if oldest pipe is far enough
- Check to see if bird has hit the floor or ceiling, update hit marker
- Draw the new bird's position
- Check to see if bird has been hit at all, and end game if so

Acknowledgements

<https://arduino.stackexchange.com/questions/42986/convert-int-to-char>

https://www.nongnu.org/avr-libc/user-manual/group_avr_eeprom.html

https://www.reddit.com/r/embedded/comments/1deyqsp/strategy_to_implement_eeprom_initialization/

<https://www.youtube.com/watch?v=NAyt5kQcn-A>